LESSON 7.7

Represent Polygons on the Coordinate Plane

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can draw polygons from their vertex coordinates on the coordinate plane, and use coordinates to find side lengths and solve perimeter and area problems.

WHY THIS MATTERS A polygon's vertices are given as coordinates: (-4.5, 3), (5, 3), (5, -3.75), and (-4.5, -3.75). What is its perimeter?

REMEMBER

Represent Polygons on the Coordinate Plane

Polygons on the Coordinate Plane Graph the points, connect them in order, and the polygon appears. Then the same coordinates that placed the corners tell you exactly how long each side is.

1. Plot all given vertices and connect them in order to form the polygon.
2. Find side lengths by counting grid units or using coordinate distance formulas.
3. Perimeter = Sum of all side lengths; Area = Use standard polygon formulas

Words: vertex *(vértice)* · polygon *(polígono)* · perimeter *(perímetro)* · distance between vertices *(distancia entre vértices)* · scale *(escala)*

PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort each pair of vertices by the kind of side it makes.

Groups: Vertical side (same x) · Horizontal side (same y)

_____ (5, 3) and (5, −3)

_____ (−2, 1) and (−2, 6)

_____ (4, −1) and (4, 7)

_____ (−4, 3) and (5, 3)

_____ (0, −2) and (6, −2)

_____ (−3, 5) and (2, 5)

HINT Look at which coordinate REPEATS in the pair — that is the direction that does not change.

2 **Complete the table. Show how you found each value.**

Find the distance between each pair of vertices.

Vertices	Shared coordinate	Distance (units)
(−4, 3) and (5, 3)		
(5, 3) and (5, −4)		
(−2, −1) and (6, −1)		

HINT Distance across zero adds the two distances from the axis: from −4 to 5 is 4 + 5.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

Vertices (−2, 6) and (−2, −6): the distance between them is

(A) 0 units
(B) 4 units
(C) 12 units
(D) 36 units

HINT Which coordinate differs?

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

What is the distance between (−5, 2) and (3, 2)?

(A) 2 units
(B) 8 units
(C) −8 units
(D) 15 units

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

What is the horizontal distance between (−4, 5) and (2, 5)?

(A) 2
(B) −2
(C) 6
(D) −6

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

What is the vertical distance between (3, −2) and (3, 6)?

(A) 3
(B) 4
(C) 8
(D) −8

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Take a side you both put under 'Slanted.' Why does subtracting coordinates give the wrong length there, and what has to be true about two vertices for subtraction to work?

SENTENCE STARTER Subtracting works when the two vertices share ___, so for ___ I cannot ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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WHY THIS MATTERS A polygon's vertices are given as coordinates: (-4.5, 3), (5, 3), (5, -3.75), and (-4.5, -3.75). What is its perimeter?

1 **Circle the letter of the best answer. Show how you know.**

How does absolute value help you find the distance between (0, −4) and (0, 6)?

(A) Subtract |−4| from |6| to get 2 units
(B) Add |−4| and |6| to get 10 units
(C) Multiply |−4| by |6| to get 24 units
(D) Absolute value is not used for distance

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Designing the Town Square: a rectangular park will have an area of 240,000 square feet. Two vertices are already marked on the planning grid, 3 units apart, and each grid unit represents 100 feet. The two known vertices of the park are 3 units apart, and each grid unit represents 100 feet. What is the park's width in feet?

(A) 3 feet
(B) 30 feet
(C) 103 feet
(D) 300 feet

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

3 **Circle the letter of the best answer. Show how you know.**

Designing the Town Square: a rectangular park will have an area of 240,000 square feet. Two vertices are already marked on the planning grid, 3 units apart, and each grid unit represents 100 feet. The park's area must be 240,000 sq ft and its width is 300 ft. What is its length?

(A) 720 feet
(B) 800 feet
(C) 80,000 feet
(D) 240,300 feet

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

4 **Circle the letter of the best answer. Show how you know.**

Designing the Town Square: a rectangular park will have an area of 240,000 square feet. Two vertices are already marked on the planning grid, 3 units apart, and each grid unit represents 100 feet. The park's length is 800 feet, and each grid unit represents 100 feet. How many units below the known vertices do the new corners sit?

(A) 0.8 units
(B) 8 units
(C) 80 units
(D) 800 units

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

5 **Circle the letter of the best answer. Show how you know.**

A rectangle has vertices (-1, 4), (5, 4), (5, 1), (-1, 1). What is its perimeter?

(A) 6 units

(B) 9 units

(C) 18 units

(D) 18 square units

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

EXPLAIN YOUR THINKING

Take a side you both put under 'Slanted.' Why does subtracting coordinates give the wrong length there, and what has to be true about two vertices for subtraction to work?

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